Code No: 121AL **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, May/June - 2019
MATHEMATICAL METHODS
(Common to EEE, ECE, CSE, EIE, IT)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART- A

(25 Marks)

1.a) If $\sum x = 10$, $\sum x^2 = 26$, $\sum y = 15$, $\sum xy = 18$, $n = 6$, find the values a and b for best fit of a straight line $y = a + bx$. [2]

b) Find the missing value from the following data:

x	1	3	5
y	-	9	26

[3]

c) How many number of subintervals are required for applying trapezoidal and Simpson rules to evaluate an integral. [2]

d) Derive an iterative formula for reciprocal of a non-zero number using Newton's Raphson method. [3]

e) Find the Fourier sine series of $f(x) = x^2$ in $[0, \pi]$. [2]

f) Find the Fourier cosine Transform of $f(x) = e^{-ax}$, a > 0. [3]

g) Solve the partial differential equation $p + q = 1$. [2]

h) Solve $\frac{\partial u}{\partial t} = 4\frac{\partial u}{\partial x} + u$ by method of separation of variables. [3]

i) Define solenoidal and irrotational vector. [2]

j) Find $Curl(\bar{r})$ where $\bar{r} = xi + yj + zk$. [3]

PART-B

(50 Marks)

2.a) Prove that n^{th} difference of a polynomial of degree n is constant and higher order differences are zero.

b) Fit a straight line of the following data

x	1	3	5	7	9
y	1.5	2.8	4.0	4.7	6.0

[5+5]

OR

3.a) Interpolate by means of Gauss backward formula, the population of town for the year 1974, given that

Year	1939	1949	1959	1969	1979	1989
Population (000)	12	15	20	27	39	52

b) Prove that (i) $\mu = \sqrt{1 + \frac{\delta^2}{4}}$ and (ii) $E = e^{hD}$, where h is step size, D is differentiable operator. [5+5]

4. Solve by Jacobi iteration method for the system of equations
$8x-3y+2z=20;\ 6x+3y+12z=35$ and $4x+11y-z=33$. [10]

OR

5. Evaluate $\int_0^2 e^{-x^2}\,dx$ using Trapezoidal rule as well as Simpson's rule, taking step size h=0.2. [10]

6. Find the Fourier series of f(x) = $\begin{cases} 0\ , & -\pi \le x \le 0 \\ \sin x, & 0 \le x \le \pi \end{cases}$; Hence show that $\frac{1}{1.3}+\frac{1}{3.5}+\frac{1}{5.7}+\ldots\ldots\ldots=\frac{1}{4}(\pi-2)$. [10]

OR

7. Find the Fourier sine and cosine transform of $f(x)=\frac{e^{-ax}}{x}$, and hence show that $\int_0^\infty \frac{e^{ax}-e^{-ax}}{x}\sin sx\,dx = \text{Tan}^{-1}\left(\frac{s}{a}\right)-\text{Tan}^{-1}\left(\frac{s}{b}\right)$, a, b > 0. [10]

8.a) Solve $p^2-q^2=x-y$.

b) Solve $x^2(y-z)p+y^2(z-x)q=z^2(x-y)$. [5+5]

OR

9. If a string of length l is initially at rest in equilibrium position and each of its points is given the velocity $v_0\sin^3\left(\frac{\pi x}{l}\right)$, find the displacement $y(x,t)$. [10]

10.a) Prove that $\text{div}(r^n\bar{r})=(n+3)r^n$, where $r=|\bar{r}|=\sqrt{x^2+y^2+z^2}$.

b) Calculate the work done by a force F= 3xyi - y^2j in moving a particle in xy-plane from (0, 1) to (1, 2) along the parabola y = x^2. [5+5]

OR

11. Verify Stokes theorem for F=(x^2+y^2)i-2xyj taken around a rectangle bounded by the lines x = a, x= -a, y = 0 and y = b. [10]

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